# Methodology and verification: Free Evidence

Companion to **When Acting on Knowledge Makes Free Evidence Harmful: Good's Theorem in Normality Models**, version 5.

## Argument and source comparison

The essay defines the value of a question using the current knowledge-conditioned probabilities and the later KDT policy. Its finite proofs identify the boundary for universally safe questions in statist, comparable models with K-normality, then analyze three specified conditional evaluations of inquiry.

The source comparison covers Goodman and Salow's normality framework, the Bjorn discussion in Belief Revision Normalized, Salow's conditional dominance principle, and the definitions and relevant equivalence results in Deference Done Better. The latter already characterizes general decision-theoretic deference. The essay's claim is the normality-specific structural characterization, not a new general criterion for valuable information.

## Verification

The finite arguments and explicit counterexamples were inspected. Exact arithmetic gives, in the five-state example, KDT value $-1952/6125$, VER value $-12847/1568000$, and ANS value $32917/217000$. The revised text explicitly fixes $s_0 = s_1$.

In the four-state example without relative defeat, KDT and ANS each value the question at $1/20$, whereas VER gives $-49/40$ and CEU gives $-49/20$. These are values of the specified inquiry evaluations with the stated later policy.

The Bjorn numerical integration was rerun for the reported payoff ratios. At $c = 1.96$, the ten-to-one KDT value is approximately $-0.0497$, while ANS's maximum price is approximately $0.0187$. A separate Gaussian calculation gives probability $0.416873$ of losing the old weight-interval belief and $0.366877$ of losing it with a discovery consistent with the old beliefs. The latter is the event described in BRN's note 18.

The supporting finite enumeration's domain is nontrivial partitions. Its random utility samples are not a verification of every utility function; the written proofs carry those universal claims. The complete 64,004-pair enumeration was not rerun for this edition. The price argument now explicitly chooses $0 < \kappa < L$ as well as a price below the positive ANS gain.

## Scope and remaining limitations

The results hold under the stated assumptions about normality, available questions, a fixed current ken and the subsequent decision policy. VER and ANS are inquiry evaluations here, not fully recursive decision theories. The analysis does not exclude every conditional repair, stake-sensitive knowledge, non-statist models, or commitment to ignore later evidence. The question-relative model of repeated measurement remains an uncomputed alternative discussed in the essay.

Proof checking and numerical work are not independent expert review or a certification of novelty.

## Matching files

The manuscript and anonymous PDF share the version 5 text. The source count is **5,739 words**, including title, abstract, headings, mathematical tokens and appendix; only the bibliography is excluded. This report records scientific methods and verification, not a complete production history or eligibility determination.